Computer Vision Project

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Project aim

This project is, first and foremost, a learning exercise: as computer-vision beginners, we wanted to try out several of the modern computer-vision techniques we studied in class, brought together as one coherent video pipeline rather than as separate isolated experiments.

Concretely, this project is to automatically detect objects in traffic videos and explain the detected objects and traffic scenes. The annotated videos could be used as demonstration material for self-driving car marketing by highlighting surrounding vehicles, traffic lights, and pedestrians, etc. The objective is to help increase customer confidence in self-driving technologies in an intuitive way. The pipeline is as planned:

- detects relevant traffic objects (vehicles, pedestrians, traffic lights, …),
- optionally tags individual cars with a brand label when the classifier is confident enough,
- generate audience-friendly captions on video (traffic scene),
- provide a simple Q&A interface for selected timestamps.

The intended use is educational and explainer contexts — for example, showing how several computer-vision components can together describe what is happening on the road in a way an everyday viewer can follow.

Data

- Self-driving videos with YOLO-compatible annotations — a public Roboflow dataset of annotated traffic scenes covering the relevant object classes (vehicles, pedestrians, traffic lights, …).
- Car brand dataset — a public Kaggle dataset of car images grouped by brand, used to train the brand classifier.

Approach

To realise this use case we combine three families of pretrained computer-vision models: an object detector for traffic objects, a pretrained vision encoder used as a feature extractor for car-brand classification, and a vision-language model for scene narration and frame-level question answering.

Concretely, we use YOLO for detection, OpenCLIP for brand classification, and a local vision-language model (Qwen3-VL via Ollama) for narration and Q&A. A working prototype across these stages is already implemented and runs end-to-end on a dashcam clip. Further methodological choices, comparisons and quantitative evaluation are reserved for the final report.

Example outcome

A successful run produces a single annotated dashcam clip that a non-technical viewer can follow end-to-end. Vehicles, pedestrians and traffic lights are boxed and labelled in real time; confidently recognised cars carry a brand tag (e.g. *Toyota*, *BMW*), while less certain ones stay as a generic *car*. A short caption in the corner of the frame describes the current scene in plain language — for example, *"a car is waiting at a red light while pedestrians cross"* — and refreshes only when the scene meaningfully changes, keeping the overlay calm and readable. After watching, the viewer can pause at any moment and ask a question about that frame (*"what colour is the traffic light?"*, *"how many cars are visible?"*) and receive a short natural-language answer. The intended takeaway is an intuitive, marketing-style demonstration of how multiple computer-vision components can jointly describe a driving scene to a general audience.